

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 10:23:24

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

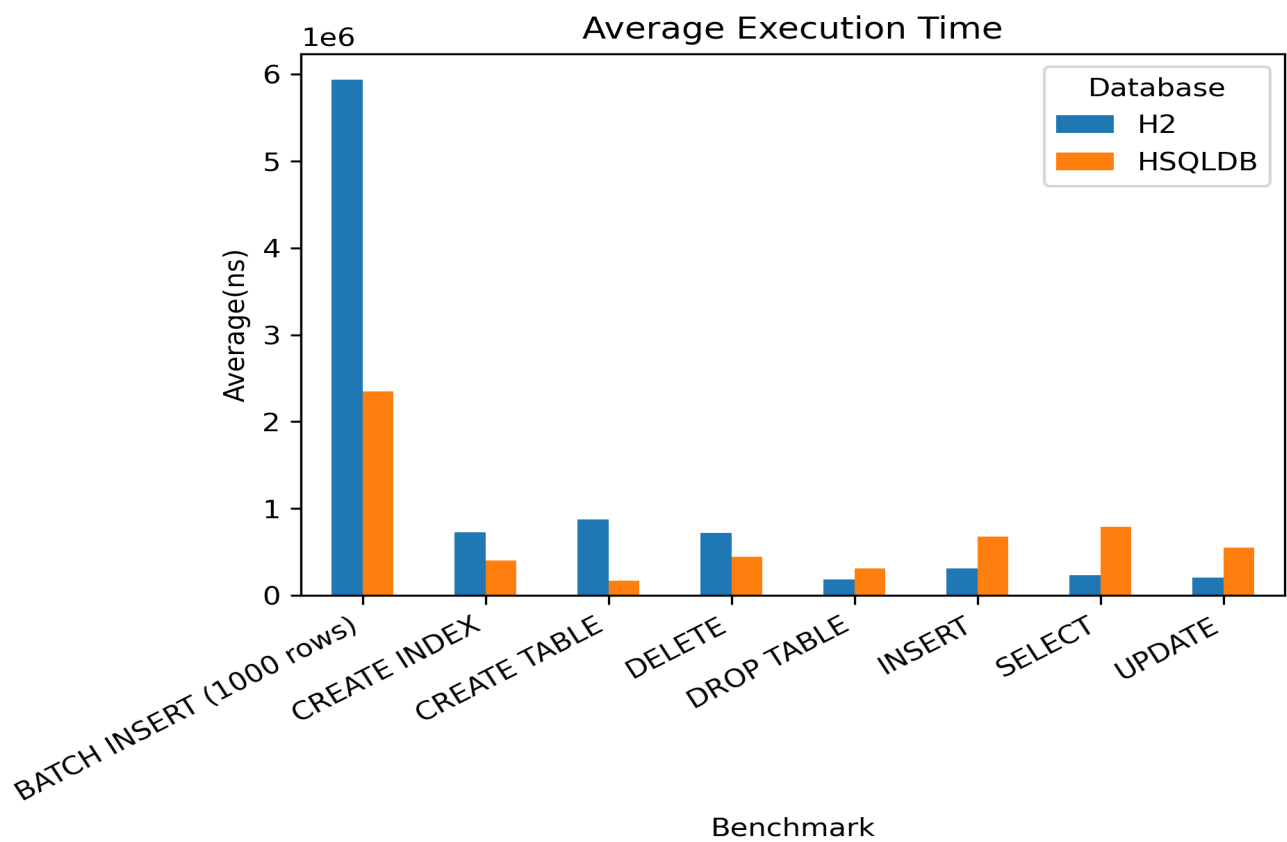
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

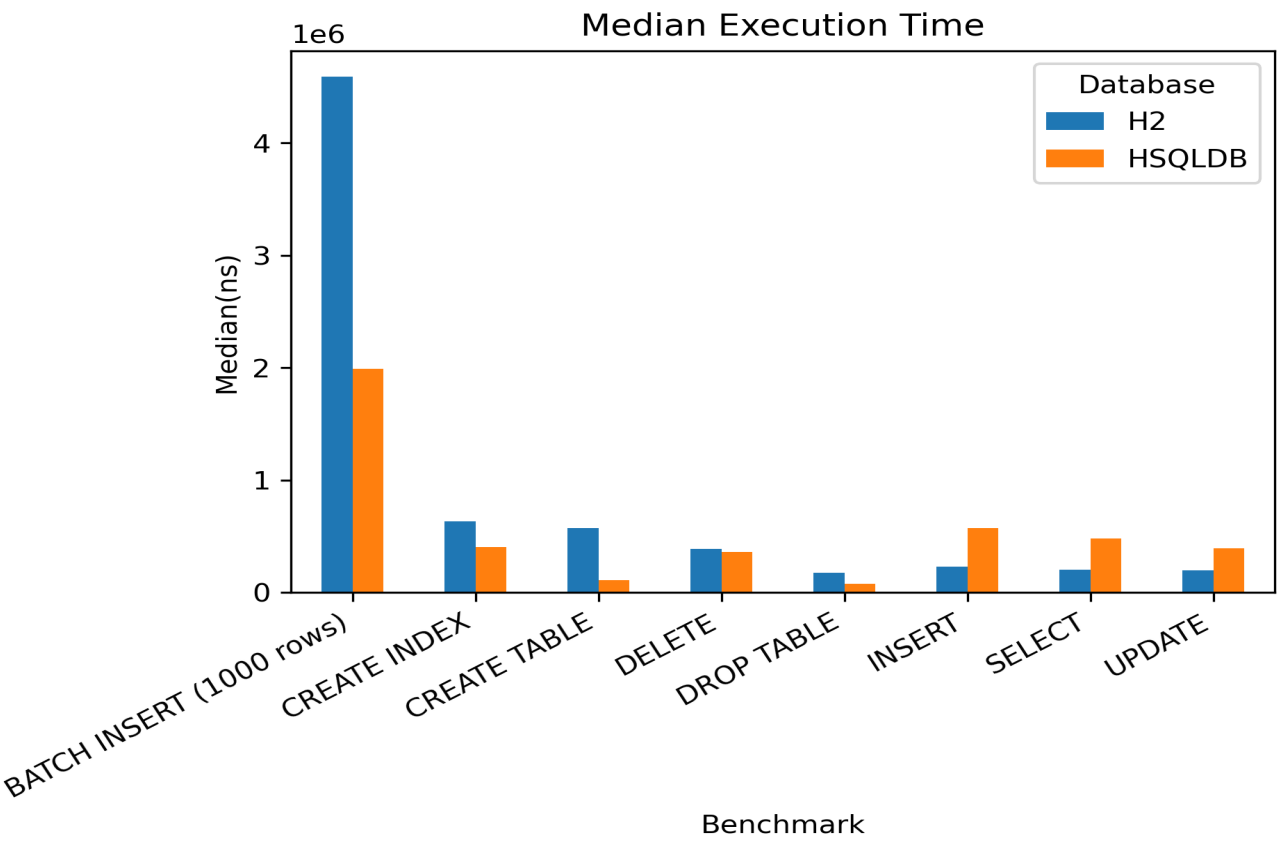
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	875541	572735	617354	291102	2622419
H2	INSERT	307883	227136	192949	156917	863205
H2	BATCH INSERT (1000 rows)	5935243	4589917	2719452	3333094	14066200
H2	SELECT	235744	200385	93259	164352	579197
H2	UPDATE	205136	196142	50770	149513	342950
H2	DELETE	722187	385671	809351	325737	3556057
H2	DROP TABLE	181437	173980	34531	147149	286713
H2	CREATE INDEX	729073	635043	302974	539302	1736319
HSQldb	CREATE TABLE	171580	108366	241411	92435	1218378
HSQldb	INSERT	677160	572019	346454	268539	1632794
HSQldb	BATCH INSERT (1000 rows)	2351377	1991218	891989	1626882	5405872
HSQldb	SELECT	792890	480039	672922	337540	3319951
HSQldb	UPDATE	547602	391958	324512	265152	1562430
HSQldb	DELETE	446150	361235	393435	240976	2133472
HSQldb	DROP TABLE	314396	77823	672695	71446	2429122
HSQldb	CREATE INDEX	404439	404968	54438	334935	544361

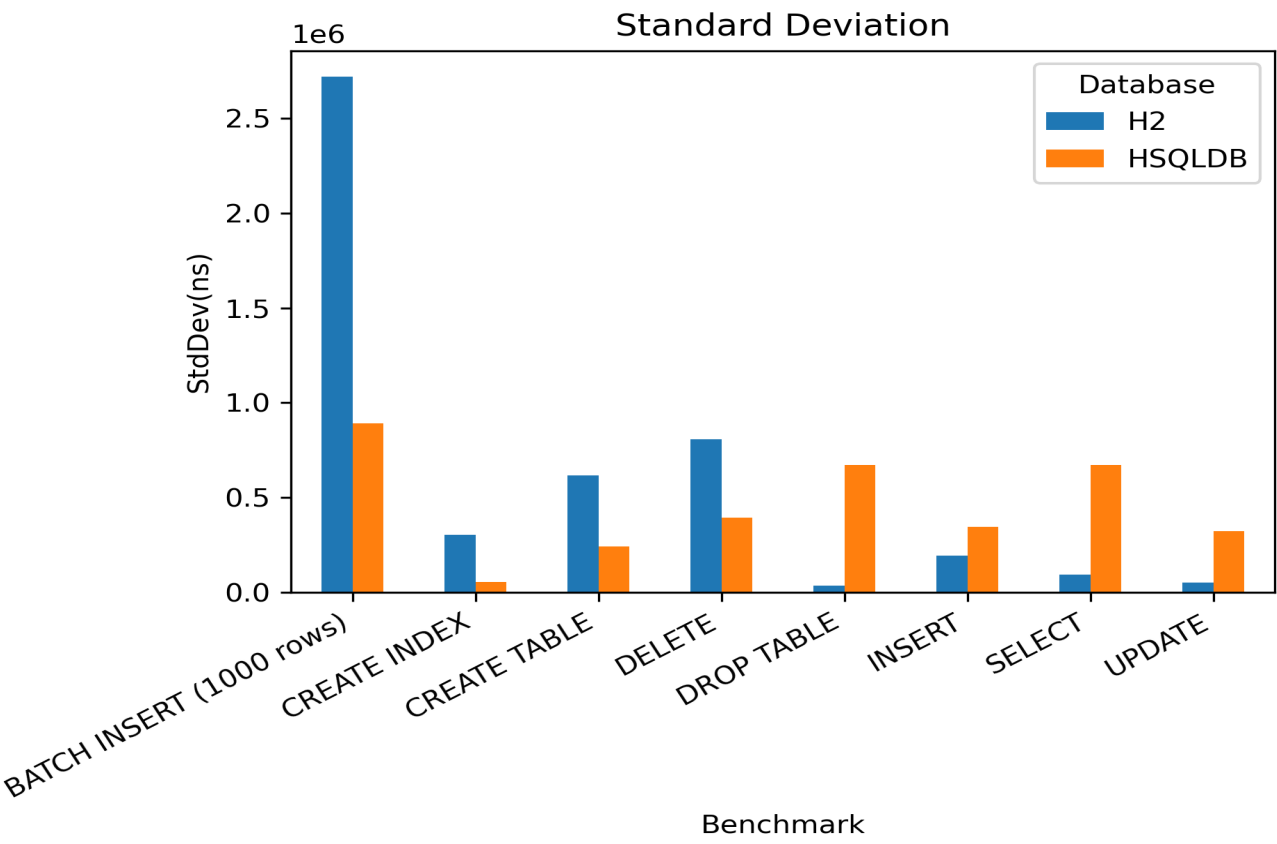
Average Execution Time



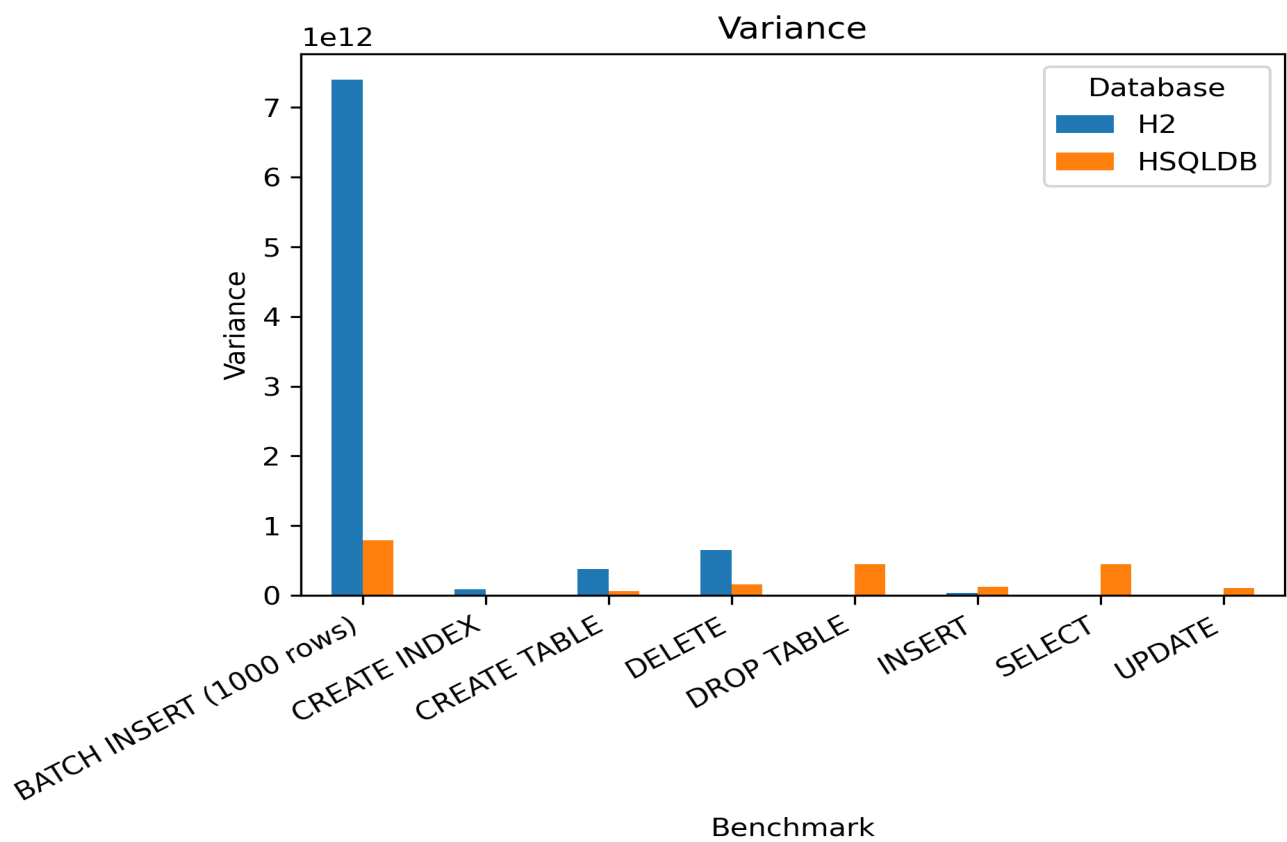
Median Execution Time



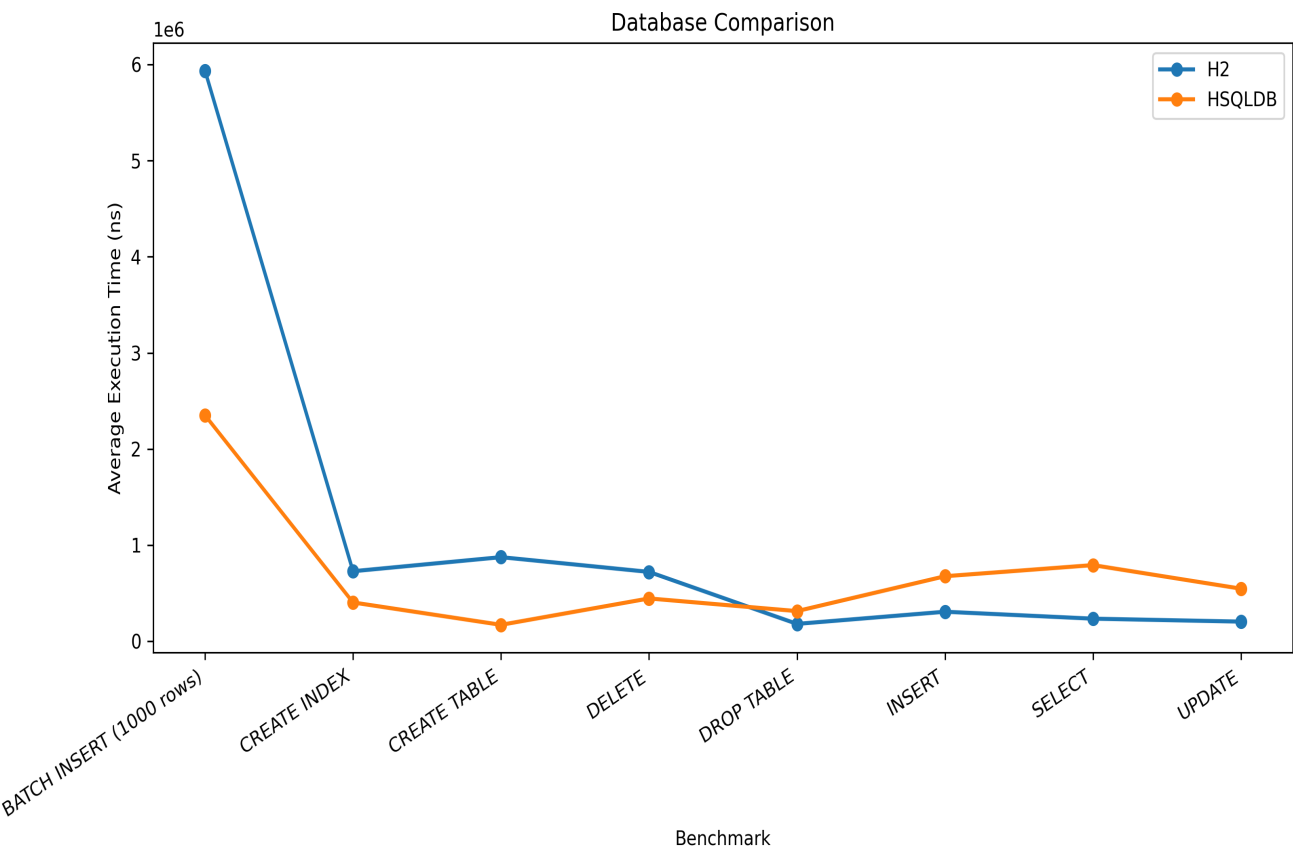
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	HSQLDB	2,351,377
CREATE INDEX	HSQLDB	404,439
CREATE TABLE	HSQLDB	171,580
DELETE	HSQLDB	446,150
DROP TABLE	H2	181,437
INSERT	H2	307,883
SELECT	H2	235,744
UPDATE	H2	205,136

Overall Benchmark Score

Database	Benchmarks Won
HSQLDB	4
H2	4

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **HSQldb** achieved the best overall performance, winning 4 benchmark categories.

The fastest benchmark measured was **CREATE TABLE** executed on **HSQldb** with an average execution time of **171,580 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **5,935,243 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.